Checking and further securing understanding of the commutative law

1 A *prime* number is an integer greater than one with exactly two factors. Fill in the blank

2 Which of the following products only use prime numbers? Tick **2** correct answers

- [ ] $4 \times 5 \times 13$
- [x] $2 \times 3 \times 19$
- [ ] $9 \times 11 \times 13$
- [x] $11 \times 13 \times 17$
- [ ] $11 \times 13 \times 15 \times 19$

3 Write 200 as a product of its prime factors. Tick **1** correct answer

- [ ] $2^2 \times 5^2$
- [ ] $2^3 \times 5$
- [x] $2^3 \times 5^2$
- [ ] $2 \times 4 \times 5^2$

4 Use the fact that $60 = 2^2 \times 3 \times 5$ to write 180 as a product of its prime factors. Tick **1** correct answer

- [ ] $2^3 \times 3 \times 5$
- [ ] $2^2 \times 3 \times 5^2$
- [x] $2^2 \times 3^2 \times 5$
- [ ] $2^2 \times 3^3 \times 5$

5 Use the fact that $60 = 2^2 \times 3 \times 5$ to write 20 as a product of its prime factors. Tick **1** correct answer

- ☐ $2^2 \times 3 \times 5 \div 3$
- ☐ $2 \times 3 \times 5$
- ☐ $2^2 \times 3$
- ☑ $2^2 \times 5$

6 $165 = 3 \times 5 \times 11$ and $42 = 2 \times 3 \times 7$. Which of the following is the most useful arrangement to calculate 165×42? Tick **1** correct answer

- ☐ $3 \times 5 \times 11 \times 2 \times 3 \times 7$
- ☐ $2 \times 3 \times 3 \times 5 \times 7 \times 11$
- ☐ $2 \times 11 \times 5 \times 3 \times 3 \times 7$
- ☑ $2 \times 5 \times 3 \times 3 \times 7 \times 11$